
**DR. A.P.J. ABDUL KALAM TECHNICAL UNIVERSITY
UTTAR PRADESH, LUCKNOW**



Evaluation Scheme & Syllabus

for

**B.Tech. Honors Degree
in**

Cyber Security

**Eligible Branches-
Computer Science and Engineering and allied Branches**

As per NEP 2020

(Effective from the Session: 2023-24)

STUDY & EVALUATION SCHEME
Honors Degree in Cyber Security
Eligible Branches- Computer Science and Engineering
& allied Branches

EVALUATION SCHEME
(Effective from the Session:2023-24)

Sl. No.	Semester	Subject Codes	Subject	Periods			Theory Sessional (TS)			Practical Sessional	Theory End Semester Examination	Practical End Semester Examination	Total	Credits
				L	T	P	CT	TA	CT+TA	PS	TE	PE		
1	IV	HTCS401	Information Theory for Cyber Security	3	1	0	30	20	50	--	100	--	150	4
2	V	HTCS501	Data Encryption and Compression	3	1	0	30	20	50	--	100	--	150	4
3	VI	HTCS601	Security Assessment and Risk Analysis	3	1	0	30	20	50	--	100	--	150	4
4	VII	HTCS701	Database Security and Access Control	3	1	0	30	20	50	--	100	--	150	4
5	VIII	HTCS801	Cyber Security Lab	0	0	4	--	--	--	50	--	50	100	2
			Total										700	18

Syllabus

HTCS401 Information Theory for Cyber Security		
Course Outcome (CO)		
At the end of course , the student will be able to understand		
CO 1	To provide an insight to information coding techniques, error correction mechanism for cyber security.	
CO 2	To introduce the principles and applications of information theory.	
CO 3	To justify how information is measured in terms of probability and entropy.	
CO 4	To learn coding schemes, including error correcting codes.	
DETAILED SYLLABUS		
Unit	Topic	Proposed Lecture
I	Shannon's foundation of Information theory, Random variables, Probability distribution factors, Uncertainty/entropy information measures, Leakage, Quantifying Leakage and Partitions, Lower bounds on key size: secrecy, authentication and secret sharing. provable security, computationally-secure, symmetric cipher.	08
II	Secrecy, Authentication, Secret sharing, Optimistic results on perfect secrecy, Secret key agreement, Unconditional Security, Quantum Cryptography, Randomized Ciphers, Types of codes: block codes, Hamming and Lee metrics, description of linear block codes, parity check Codes, cyclic code, Masking techniques.	08
III	Information-theoretic security and cryptograph, basic introduction to Diffie-Hellman, AES, and side-channel attacks.	08
IV	Secrecy metrics: strong, weak, semantic security, partial secrecy, Secure source coding: rate-distortion theory for secrecy systems, side information at receivers, Differential privacy, Distributed channel synthesis.	08
V	Digital and network forensics, Public Key Infrastructure, Light weight cryptography, Elliptic Curve Cryptography and applications	08
Text books: 1. Information Theory and Coding, Muralidhar Kulkarni, K S Shivaprakasha, John Wiley & Sons. 2. Communication Systems: Analog and digital, Singh and Sapre, Tata McGraw Hill. 3. Fundamentals in information theory and coding, Monica Borda, Springer. 4. Information Theory, Coding and Cryptography R Bose.		

HTCS501		Data Encryption and Compression	
Course Outcome (CO)			
At the end of course , the student will be able to understand			
CO 1	At the end of this course the student will have the knowledge of Encryption Techniques.		
CO 2	To introduce the principles and applications of Cryptography		
CO 3	To understand the concept of the Data Compression.		
CO 4	To introduce the principles and applications of Entropy encoding		
DETAILED SYLLABUS			
Unit	Topic		Proposed Lecture
I	Introduction to Security: Need for security, Security approaches, Principles of security, Types of attacks. Encryption Techniques: Plaintext, Cipher text, Substitution & Transposition techniques, Encryption & Decryption, Types of attacks, Key range & Size.		08
II	Symmetric & Asymmetric Key Cryptography: Algorithm types & Modes, DES, IDEA, Differential & Linear Cryptanalysis, RSA, Symmetric & Asymmetric key together, Digital signature, Knapsack algorithm.		08
III	Case Studies of Cryptography: Denial of service attacks, IP spoofing attacks, Conventional Encryption and Message Confidentiality, Conventional Encryption Algorithms, Key Distribution. Public Key Cryptography and Message Authentication: Approaches to Message Authentication, SHA-1, MD5, Public-Key Cryptography Principles, RSA, Digital, Signatures, Key Management, Firewall.		08
IV	Introduction: Need for data compression, Fundamental concept of data compression & coding, Communication model, Compression ratio, Requirements of data compression, Classification. Methods of Data Compression: Data compression-- Loss less & Lossy.		08
V	Entropy encoding-- Repetitive character encoding, Run length encoding, Zero/Blank encoding; Statistical encoding-- Huffman, Arithmetic & Lempel-Ziv coding; Source encoding-- Vector quantization (Simple vector quantization & with error term). Recent trends in encryption and data compression techniques.		08
Text books: 1. Cryptography and Network Security, Mohammad Amjad, John Wiley & Sons. 2. Cryptography & Network Security by Atul Kahate, TMH. 3. Information Theory and Coding, Muralidhar Kulkarni, K S Shivaprakasha, John Wiley & Sons. 4. Cryptography and Network Security by B. Forouzan, McGraw-Hill. 5. The Data Compression Book by Nelson, BPB.			

HTCS601 Security Assessment and Risk Analysis		
Course Outcome (CO)		
At the end of course , the student will be able to understand		
CO 1	To Describe the concepts of risk management in information security. Define and differentiate various Contingency Planning components.	
CO 2	To Skilled to be able to describe the escalation process from incident to disaster in case of security disaster.	
CO 3	To understand the concept of the Data Compression.	
CO 4	To Design a Disaster Recovery Plan for sustained organizational operations.	
DETAILED SYLLABUS		
Unit	Topic	Proposed Lecture
I	SECURITY BASICS: Information Security (INFOSEC) Overview: critical information characteristics – availability information states – processing security countermeasures-education, training and awareness, critical information , characteristics – confidentiality critical information characteristics – integrity, information states – storage, information states – transmission, security , countermeasures- policy, procedures and practices, threats, vulnerabilities.	08
II	Threats to and Vulnerabilities of Systems: Threats, major categories of threats (e.g., fraud, Hostile Intelligence Service (HOIS). Countermeasures: assessments (e.g., surveys, inspections). Concepts of Risk Management: consequences (e.g., corrective action, risk assessment), cost/benefit analysis and implementation of controls, monitoring the efficiency and effectiveness of controls (e.g., unauthorized or inadvertent disclosure of information).	08
III	Security Planning: directives and procedures for policy mechanism. Contingency Planning/Disaster Recovery: agency response procedures and continuity of operations, contingency plan components, determination of backup requirements, development of plans for recovery actions after a disruptive event.	08
IV	Personnel Security Practices and Procedures: access authorization/verification (need- to-know), contractors, employee clearances, position sensitivity, security training and awareness, systems maintenance personnel. Auditing and Monitoring: conducting security reviews, effectiveness of security programs, investigation of security breaches, privacy review of accountability controls, review of audit trails and logs.	08
V	Operations Security (OPSEC): OPSEC surveys/OPSEC planning INFOSEC: computer security – audit, cryptography-encryption (e.g., point-to-point, network, link). Case study of threat and vulnerability assessment.	08
Text books: 1. Information Systems Security, 2ed: Security Management, Metrics, Frameworks and Best Practices, Nina Godbole, John Wiley & Sons. 2. Principles of Incident Response and Disaster Recovery, Whitman & Mattord, Course Technology ISBN: 141883663X.		

Corresponding Online Resources:

1. Introduction to Cyber Security, https://swayam.gov.in/nd2_nou20_cs01/preview
2. (Web Link) http://www.cnss.gov/Assets/pdf/nstissi_4011.pdf

HTCS 701		Database Security and Access Control	
Course Outcome (CO)			
At the end of course , the student will be able to understand			
CO 1	To provide fundamentals of database security.		
CO 2	To Skilled to be able to describe the escalation process from incident to disaster in case of security disaster. To understand and implement classical models and algorithms.		
CO 3	To analyze the data, identify the problems, and choose the relevant models and algorithms to apply.		
CO 4	To assess the strengths and weaknesses of various access control models and to analyze their behaviour.		
DETAILED SYLLABUS			
Unit	Topic		Proposed Lecture
I	Introduction to Access Control, Purpose and fundamentals of access control.		08
II	Policies of Access Control, Models of Access Control, and Mechanisms, Discretionary Access Control (DAC), Non- Discretionary Access Control, Mandatory Access Control (MAC). Capabilities and Limitations of Access Control Mechanisms: Access Control List (ACL) and Limitations, Capability List and Limitations.		08
III	Role-Based Access Control (RBAC) and Limitations, Core RBAC, Hierarchical RBAC, Statically Constrained RBAC, Dynamically Constrained RBAC, Limitations of RBAC. Comparing RBAC to DAC and MAC Access Control policy, Integrating RBAC with enterprise IT infrastructures: RBAC for WFMSs, RBAC for UNIX and JAVA environments.		08
IV	Smart Card based Information Security, Smart card operating system-fundamentals, design and implantation principles, memory organization, smart card files, file management. PPS Security techniques- user identification, smart card security, quality assurance and testing, smart card life cycle-5 phases, smart card terminals.		08
V	Cloud Data Security: Recent trends in Database security and access control mechanisms. Cloud Data Audit: Intro, Audit, Best Practice, Key management, Cloud Key Management Audit.		08
Text books:			
1. Role Based Access Control: David F. Ferraiolo, D. Richard Kuhn, Ramaswamy Chandramouli.			
Reference Books :			
1.Database Security and Auditing, Hassan A. Afyouni, India Edition, CENGAGE Learning, 2009.			
2. Database Security, Castano, Second edition, Pearson Education.			
Corresponding Online Resources:			
1. http://www.smartcard.co.uk/tutorials/sct-itsc.pdf : Smart Card Tutorial.			
2. Advanced System Security Topics, https://www.coursera.org/lecture/advanced-system-security-topics/role-based-access-control-rbac-bYvzS .			

1. Implement the following SUBSTITUTION & TRANSPOSITION TECHNIQUES concepts:
 - a) Caesar Cipher
 - b) Playfair Cipher
 - c) Hill Cipher
 - d) Vigenere Cipher
 - e) Rail fence – row & Column Transformation
2. Implement the following algorithms
 - a) DES
 - b) RSA Algorithm
 - c) Diffie-Hellman
 - d) MD5
 - e) SHA-1
3. Implement the Signature Scheme - Digital Signature Standard
4. Demonstrate how to provide secure data storage, secure data transmission and for creating digital signatures (GnuPG)
5. Setup a honey pot and monitor the honeypot on network (KF Sensor)
6. Installation of rootkits and study about the variety of options
7. Perform wireless audit on an access point or a router and decrypt WEP and WPA. (Net Stumbler)
8. Demonstrate intrusion detection system (ids) using any tool (snort or any other s/w)
9. Implement Huffman coding.
 - 10 Implement Arithmetic Coding.
11. Implement one dimension DCT
12. Implement Chinese Remainder Theorem
13. Implement Ceaser Cipher Algorithm
14. TCP scanning using NMAP2.
15. Port scanning using NMAP3.
16. TCP / UDP connectivity using Netcat
17. Network vulnerability using OpenVAS5.
18. Web application testing using DVWA
19. Manual SQL injection using DVWA
20. XSS using DVWA
21. Automated SQL injection with SqlMap

Note: Subject Teachers may add/delete the experiments as per the subject requirements